

Grace Bianchi

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EDUCATION

University of California, Santa Barbara

B.S. Statistics and Data Science, B.A. Economics

Expected Graduation: June 2027

3.88/4.00 GPA

Honors Thesis in Economics, Data Science Capstone – Admitted, coursework begins Fall 2026

Relevant Coursework: Stochastic Processes, Probability Theory, Game Theory, Linear Algebra, Mathematical Statistics, Statistical Machine Learning, Regression Analysis, Econometrics, Vector Calculus, Differential Equations, Discrete Math

PROFESSIONAL EXPERIENCE

MUFG | Data Quality and Solutions Summer Analyst

June 2026 – August 2026

- Built a Python scraper into PostgreSQL to analyze 10 years of loss-run history for insurance umbrella risk analysis
- Benchmarked umbrella premium structure against risk profile change, peer data, and PML report to recommend a coverage limit reduction saving \$250K+
- Validated reverse-engineered financial crimes rules in SQL during a platform migration for legacy-to-new parity

SageView Advisory Group | Data and Analytics Intern

June 2025 – August 2025

- Built a client segmentation model in R using weighted z-scores to score and tier 150+ institutional clients, and applied a decision tree to generate targeted improvement recommendations based on lowest-performing metrics
- Developed a SQL and PowerBI dashboard to search and visualize nested account, plan, and contact data
- Prepared investment reports for client meetings, compiling portfolio holdings and YTD returns

University of California, Irvine Spatial Neuroscience Lab | Research Assistant

June 2022 – January 2023

- Developed Python scripts to process VR-based spatial navigation experimental data for statistical analysis
 - Supported integrating VR, fMRI, and diffusion-weighted imaging to study neural systems underlying navigation
 - Assisted in EDA and scientific literature review on hippocampal and cortical contributions to spatial cognition
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PROJECTS

Markov Regime-Switching Volatility Forecaster

- Combined GARCH/HAR-RV baselines with a BIC-selected 4-state HMM regime detector to forecast SPY volatility, validated against 5 major volatility events including COVID, Volmageddon, and 2025 tariff selloff
- Designed a regime-aware VaR framework using Markov-switching forecasts and Extreme Value Theory tail fitting, backtested via Kupiec/Christoffersen coverage tests on 887 days
- Ran Monte Carlo regime simulations to build conditional volatility distributions, using Diebold-Mariano testing to show improved tail calibration over classical HAR-RV

Stellar Classification: Using Machine Learning to Assess the Role of Redshift

- Fit 14 classification models (7 types x full/reduced feature sets) in R to classify stellar objects from SDSS data
- Demonstrated that redshift is a critical predictor – removing it drops Random Forest test accuracy from 97.9% to 84.1%, with STAR classification most impacted (39% misclassification rate without redshift)
- Tuned hyperparameters via 10-fold cross-validation; evaluated best models using ROC AUC, accuracy, ROC curves, and confusion matrices

Housing Market Segmentation using Clustering and Association Rules

- Applied k-means, hierarchical (Ward.D2), and PAM clustering to 1,047 homes in R, selecting k=3 via elbow method and validating with Hopkins statistic (H=0.999) and cophenetic correlation
 - Identified three market segments (entry-level, mid-market, high-value) with PCA confirming a dominant size-value axis explaining 49.1% of variance across 7 housing features
 - Mined 6,749 Apriori association rules, finding that a large living area and 3+ bathrooms predict an expensive price tier with 100% confidence and a lift of 2.94
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SKILLS & ACTIVITIES

Programming Languages: Python, R, SQL, Java, Stata

Tools: Microsoft Office, Power BI, Excel, Git, Jupyter, LaTeX, RStudio

Awards and Honors: UCSB Dean's Honors, AP Scholar with Distinction, AMC Math Competition Participant

Activities: Undergraduate Learning Assistant: Intermediate Microeconomic Theory (Summer 2026), Statistics for Economics (Fall 2026 - Spring 2027), UCSB Data Science Club (2024-2026), UCSB Data Science Collaborative (2024-2026), Kappa Alpha Theta (2023-2025), Scitrek (2024)

Interests: Sudoku (Extreme PR: 4 minutes 56 seconds), WWII history, Risk, Fleetwood Mac